

MaxPooling2D(pool_size=3, strides=2, padding="same")
name="pool1"

Given: $N = 14$ (input from conv1, shape $14 \times 14 \times 8$), $K = 3$, $S = 2$

“Same” padding target formula:

$$\text{out} = \left\lceil \frac{N}{S} \right\rceil$$

$$\text{out} = \left\lceil \frac{14}{2} \right\rceil = \lceil 7 \rceil = 7$$

Solve for padding P :

$$P = \frac{S \cdot (\lceil N/S \rceil - 1) - N + K}{2}$$

$$P = \frac{2 \cdot (7 - 1) - 14 + 3}{2}$$

$$P = \frac{2 \cdot 6 - 14 + 3}{2}$$

$$P = \frac{12 - 14 + 3}{2}$$

$$P = \frac{1}{2} = 0.5$$

Since P is not a whole number, padding is asymmetric: 0 on one side, 1 on the other (total $P = 1$).

Verify with general output formula:

$$\text{out} = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

$$\text{out} = \left\lfloor \frac{14 + 1 - 3}{2} \right\rfloor + 1 = \left\lfloor \frac{12}{2} \right\rfloor + 1 = 6 + 1 = 7$$

Result:

Input (14, 14, 8) $\longrightarrow$ Output (7, 7, 8)

Conv2D(8, kernel_size=3, strides=2, padding="valid")

name="conv1"

Given: $N = 28$ (input shape $28 \times 28 \times 1$), $K = 3$, $S = 2$, $P = 0$

General output formula:

$$\text{out} = \left\lfloor \frac{N + 2P - K}{S} \right\rfloor + 1$$

$$\text{out} = \left\lfloor \frac{28 + 0 - 3}{2} \right\rfloor + 1$$

$$\text{out} = \left\lfloor \frac{25}{2} \right\rfloor + 1$$

$$\text{out} = 12 + 1$$

$$\text{out} = 13$$

Result:

Input (28, 28, 1) $\longrightarrow$ Output (13, 13, 8)